

WAZUH SIEM

Deployment & Operations Guide

Agent Installation · Feature Reference · Log Monitoring Manual

STATUS: LIVE	GDPR ART. 32	18 AGENTS DEPLOYED
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Prepared By	Karthikeyan T
Wazuh Version	vx.x.x.x
Total Agents	18 Agents
Date	June 2026
Classification	Technical reference

1. Introduction

This document is the technical reference for a Wazuh Security Information and Event Management (SIEM) deployment. Wazuh version 4.14.5 was deployed across 18 endpoints covering both Linux (Ubuntu) and Windows machines.

The document is structured in three parts: (1) the step-by-step agent deployment process performed on each machine, (2) a comprehensive feature reference covering every module in Wazuh, and (3) a practical log monitoring guide showing how to use each feature for day-to-day security operations.

2. What is Wazuh?

Wazuh is a free and open-source security platform that provides unified XDR (Extended Detection and Response) and SIEM capabilities. It protects endpoints — workstations, servers, and cloud infrastructure — by collecting security telemetry, analysing it in real time, and generating alerts when threats, misconfigurations, or compliance violations are detected.

Wazuh operates on a manager–agent architecture:

- Wazuh Manager:** The central server that receives data from all agents, runs correlation rules, generates alerts, and serves the web dashboard.

- **Wazuh Agent:** A lightweight process installed on each endpoint. It collects logs, file changes, running processes, network connections, and vulnerability data and forwards them securely to the manager.
- **Wazuh Dashboard:** A web-based interface (built on OpenSearch Dashboards) used to visualise alerts, investigate events, and run compliance checks.

 **NOTE:** Wazuh agents communicate with the manager on port 1514 (UDP/TCP). Ensure this port is open in any host firewall on both manager and agent machines.

3. Wazuh Agent Deployment

3.1 Overview

Wazuh agents were installed on all 18 endpoints in the office network. The standard installation method used was the Wazuh .deb package for Ubuntu/Debian systems, and the .msi installer for Windows systems. Each agent was registered to the Wazuh Manager during installation and assigned a meaningful name matching the endpoint's role.

3.2 Agent Inventory

The following 18 agents were successfully deployed and are visible in the Wazuh dashboard:

ID	Agent Name	IP Address	OS	Status
002	endpoint-02	192.168.x.x	Ubuntu 22.04.5 LTS	active
003	server-agent	192.168.x.x	Ubuntu 24.04.3 LTS	active
004	endpoint-04	192.168.x.x	Ubuntu 20.04.6 LTS	active
005	endpoint-05	192.168.x.x	Ubuntu 24.04.3 LTS	active
006	endpoint-06	192.168.x.x	Ubuntu 22.04.5 LTS	active
007	endpoint-07	192.168.x.x	Ubuntu 20.04.6 LTS	active
008	endpoint-08	192.168.x.x	Ubuntu 22.04.5 LTS	active
009	endpoint-09	192.168.x.x	Ubuntu 22.04.5 LTS	active
010	endpoint-10	192.168.x.x	Ubuntu 22.04.5 LTS	active
011	endpoint-11	192.168.x.x	Ubuntu 20.04.6 LTS	active
012	endpoint-12	192.168.x.x	Ubuntu 22.04.5 LTS	active
013	endpoint-13	192.168.x.x	Ubuntu 20.04.6 LTS	active
015	endpoint-15	192.168.x.x	Ubuntu 24.04.2 LTS	active
017	win-agent-01	192.168.x.x	Windows 11 Pro	active
018	win-agent-02	192.168.x.x	Windows 11 Pro	active

019	win-agent-03	192.168.x.x	Windows 11 Pro	active
020	endpoint-20	192.168.x.x	Ubuntu 20.04.6 LTS	active

Note: Agent names and IP addresses have been anonymised for this public portfolio document.

4. Agent Installation Process

4.1 Linux (Ubuntu/Debian) — Step-by-Step

The following commands were executed on every Linux machine (Ubuntu 20.04, 22.04, and 24.04) to install and register the Wazuh agent. Replace `<MANAGER_IP>` with your Wazuh manager address and `<agent_name>` with the device's assigned name.

Step 1 — Download the Wazuh Agent Package

```
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.5-1_amd64.deb
```

Downloads the official Wazuh agent .deb package version 4.14.5 for 64-bit Linux systems from the Wazuh package repository.

Step 2 — Install the Agent with Manager Registration

```
sudo WAZUH_MANAGER='<MANAGER_IP>' \
  WAZUH_AGENT_NAME='<agent_name>' \
  dpkg -i ./wazuh-agent_4.14.5-1_amd64.deb
```

The two environment variables passed before `dpkg` configure the agent at install time:

- **WAZUH_MANAGER:** Sets the IP address of the Wazuh Manager the agent will connect to.
- **WAZUH_AGENT_NAME:** Sets the display name for this agent in the dashboard. Use a consistent naming convention.

i NOTE: You must run this command with `sudo`. The installation writes the manager IP into `/var/ossec/etc/ossec.conf` automatically.

Step 3 — Reload Systemd

```
sudo systemctl daemon-reload
```

Reloads systemd's unit file database to recognise the new `wazuh-agent.service` unit installed by the package.

Step 4 — Enable the Agent Service

```
sudo systemctl enable wazuh-agent
```

Enables the agent to start automatically on system boot.

Step 5 — Start the Agent Service

```
sudo systemctl start wazuh-agent
```

Starts the Wazuh agent immediately. After this command the agent will connect to the manager and appear in the Wazuh dashboard Endpoints list.

Step 6 — Verify Agent Status

```
sudo systemctl status wazuh-agent
# Expected output:
# • wazuh-agent.service - Wazuh agent
#   Active: active (running)
```

Verify in Dashboard: After starting the agent, wait 30–60 seconds then refresh Wazuh Dashboard > Endpoints. The new agent should appear with status 'active' (green dot).

4.2 Windows — Step-by-Step

For Windows 11 machines, the Wazuh MSI installer was used via PowerShell.

Step 1 — Download the Windows Installer

```
# Run in PowerShell (Administrator)
Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.5-1.msi `
-OutFile wazuh-agent-4.14.5-1.msi
```

Step 2 — Install with Manager Registration

```
msiexec.exe /i wazuh-agent-4.14.5-1.msi /q `
WAZUH_MANAGER='<MANAGER_IP>' `
WAZUH_AGENT_NAME='<agent_name>'
```

Step 3 — Start the Service

```
NET START WazuhSvc
# Or via PowerShell:
Start-Service -Name WazuhSvc
```

i NOTE: On Windows, the agent configuration file is located at: C:\Program Files (x86)\ossec-agent\ossec.conf

4.3 Verifying Registration from the Manager

```
# List all registered agents:
sudo /var/ossec/bin/agent_control -l
```

```
# Check a specific agent (e.g. ID 009):  
sudo /var/ossec/bin/agent_control -i 009
```

5. Wazuh Dashboard — Overview

The Wazuh Dashboard is the main interface for security monitoring, accessible via the manager's web UI using administrator credentials. The Overview page provides a real-time summary of all security events across all managed endpoints.

5.1 Alert Severity Levels

The Wazuh overview shows a 24-hour alert summary across four severity levels based on rule levels:

Critical (15+)	High (12–14)	Medium (7–11)	Low (0–6)
0	11	3,445	20,667

Sample alert counts from a monitored endpoint during a 24-hour window — illustrates the volume and distribution of events Wazuh surfaces.

5.2 Individual Agent Detail

Each endpoint has a dedicated detail page showing:

- **Vulnerability Detection:** Critical / High / Medium / Low CVE counts with affected package names
- **MITRE ATT&CK:** Top tactics mapped from detected events (Defense Evasion, Privilege Escalation, Initial Access, Persistence)
- **CIS Benchmark:** Security Configuration Assessment score against the relevant CIS benchmark (e.g. CIS Ubuntu Linux 22.04 v2.0)
- **System Inventory:** Hardware specs, OS version, running processes, open ports, and installed packages

6. Wazuh Feature Reference

Wazuh provides a comprehensive set of security capabilities accessible through its dashboard. Each feature is described below with guidance on access and usage.

6.1 Threat Hunting

Proactively search security events and logs across all agents to identify hidden threats. Threat Hunting allows security analysts to query the entire event database for indicators of compromise (IoCs) or suspicious patterns that automated rules may not have flagged.

How to Use

- **Navigate to:** Wazuh Dashboard > Threat Hunting
- **Time Range:** Set the time window using the date picker (top right). Start with 'Last 24 hours' for investigations.
- **Search Filters:** Use the search bar to filter events. Example: `data.win.eventdata.logonType: 10` finds remote logon events on Windows.
- **Add Columns:** Click 'Add filter' to add specific fields like `agent.name`, `rule.description`, `data.srcip`.
- **Investigate:** Click any event row to expand it and see the full JSON payload.

Useful Queries

```
# Failed SSH logins in last 24h
rule.id: 5710

# Privilege escalation events
rule.mitre.tactic: "Privilege Escalation"

# Events from a specific agent
agent.name: endpoint-09

# High/critical severity only
rule.level: [12 TO 15]
```

6.2 File Integrity Monitoring (FIM)

Alerts on any creation, modification, deletion, or permission change to monitored files and directories. FIM is one of the most critical Wazuh capabilities for detecting insider threats, malware persistence, and unauthorised changes.

How to Configure FIM

```
# On Ubuntu: /var/ossec/etc/ossec.conf
# Add inside <syscheck> section:
<directories check_all="yes" report_changes="yes">/etc</directories>
<directories check_all="yes" report_changes="yes">/usr/bin</directories>
<directories check_all="yes" report_changes="yes">/home</directories>

# Restart agent after config change:
sudo systemctl restart wazuh-agent
```

How to Use the FIM Dashboard

- **Navigate to:** Endpoints > [Select Agent] > File Integrity Monitoring
- **Events Table:** Shows all file events: who changed what file, when, and what changed (added, modified, deleted)
- **Inventory:** FIM > Inventory tab shows the current baseline snapshot of monitored files

Best Practice: Configure FIM on `/etc`, `/var/ossec`, `/home`, and any application configuration directories to detect if any team member modifies system files.

6.3 Security Configuration Assessment (SCA)

Audits endpoint configurations against security benchmarks like CIS, PCI DSS, and NIST. SCA automatically scans endpoints against industry-standard benchmarks and generates pass/fail reports.

How to Use

- **Navigate to:** Endpoints > [Select Agent] > Configuration Assessment
- **Policy List:** See all benchmark policies applied to the endpoint (e.g. CIS Ubuntu Linux 22.04 Benchmark v2.0.0)
- **Drill Down:** Click a policy to see each individual check — passed (green), failed (red), or not applicable
- **Details:** Each failed check explains the requirement, rationale, remediation steps, and relevant NIST/CIS control
- **Score:** $\text{Score (\%)} = \frac{\text{passed}}{\text{passed} + \text{failed}} \times 100$. Target > 80% for a well-hardened system.

Remediation Workflow

- Open Configuration Assessment for an agent and filter by 'Failed' checks
- Click a failed check to read the remediation steps
- Apply the fix on the endpoint (e.g. disable root SSH login)
- Restart the agent: `sudo systemctl restart wazuh-agent` to trigger a re-scan
- Re-check the dashboard — the check should now show as Passed

i NOTE: SCA scans run on a schedule (default every 12 hours). To trigger an immediate scan, restart the agent: `sudo systemctl restart wazuh-agent`

6.4 Malware Detection

Detects indicators of compromise (IoCs) using rootcheck, FIM anomaly detection, and optional integration with VirusTotal. Wazuh uses multiple complementary mechanisms: rootcheck scans for rootkits and hidden processes, FIM detects file changes consistent with malware installation.

Rootcheck

- **What it does:** Scans for rootkits, trojan processes, suspicious ports, and hidden files at scheduled intervals
- **Alerts:** Rootkit detection events appear with rule IDs in the 510xx range

VirusTotal Integration (Optional)

Configure VirusTotal integration in `ossec.conf` on the manager to automatically submit file hashes from FIM events:

```
# In /var/ossec/etc/ossec.conf on the MANAGER:
<integration>
  <name>virustotal</name>
```

```
<api_key>YOUR_VIRUSTOTAL_API_KEY</api_key>
<rule_id>554,550</rule_id>
<alert_format>json</alert_format>
</integration>
```

How to Use

- **Navigate to:** Wazuh Dashboard > Overview (agent level) > Malware Detection tile
- **Check FIM alerts:** Rule IDs 550–561 indicate file modifications that may indicate malware activity
- **Investigate:** Expand any alert to see the full file path, old and new MD5/SHA hash, and the user who made the change

6.5 Vulnerability Detection

Continuously scans installed software packages against the NVD CVE database to identify known vulnerabilities. Wazuh agents collect installed package lists and send them to the manager, which compares them against the National Vulnerability Database (NVD) to identify CVEs.

How to Use

- **Navigate to:** Endpoints > [Agent] > Vulnerability Detection
- **Summary:** See counts of Critical / High / Medium / Low vulnerabilities with package names
- **Filter:** Filter by severity, package name, or CVE ID
- **CVE Detail:** Click any vulnerability to see: CVE ID, CVSS score, affected package version, fixed version (if available)
- **Top 5 Packages:** Shows which packages carry the most associated CVEs — highest priority for patching

Remediation for Ubuntu Agents

```
sudo apt-get update
sudo apt-get upgrade linux-image-generic
sudo reboot
```

```
# After reboot, verify running kernel version:
uname -r
```

Priority: Focus on Critical (CVSS 9.0+) and High (CVSS 7.0–8.9) vulnerabilities first. Commonly exploited packages like `imagemagick` should be patched immediately.

6.6 MITRE ATT&CK

Maps detected security events to the MITRE ATT&CK framework's tactics, techniques, and procedures (TTPs). The MITRE ATT&CK integration automatically tags security alerts with the corresponding ATT&CK tactic and technique, allowing the security team to understand the attacker's intent.

Top Tactics Observed (Sample — 24h Window)

Tactic	Count	Description
Defense Evasion	14	Attacker actions to avoid detection (clearing logs, disabling tools)
Privilege Escalation	13	Attempts to gain higher access rights (sudo abuse, SUID exploitation)
Initial Access	12	Techniques used to gain first foothold (phishing, exploitation)
Persistence	12	Maintaining access after compromise (cron jobs, startup scripts)
Lateral Movement	1	Moving from one system to others in the network

How to Use

- **Navigate to:** Endpoints > [Agent] > MITRE ATT&CK
- **Tactics Heatmap:** Click any tactic tile to see all alerts tagged with that tactic
- **Technique Drilldown:** Click a technique to see individual events — exact log, source IP, user, and command
- **Network-wide View:** The heatmap at the overview level shows which tactics have been observed across all agents

6.7 Threat Intelligence

Enriches alerts with external threat intelligence data to identify known malicious IPs, domains, and file hashes. Wazuh integrates with threat intelligence feeds to automatically enrich security events.

Available Modules

- **Vulnerability Detection:** Cross-references packages with the NVD CVE database
- **VirusTotal:** File hash lookups for malware identification (via API integration)
- **AlienVault OTX:** Configure OSSEC integration to pull indicators from AlienVault OTX pulse feeds

How to Use

- **Navigate to:** Dashboard > Overview > Threat Intelligence panel
- **Threat Hunting:** Click to open the full event browser filtered by threat-intelligence-enriched events
- **MITRE ATT&CK:** Click to see all events mapped to ATT&CK techniques across all agents

6.8 IT Hygiene

Continuously assesses system, software, process, and network layers for misconfigurations and anomalies. IT Hygiene provides an inventory of each endpoint's current state — essential for baseline validation and detecting unexpected changes.

How to Use

- **Navigate to:** Endpoints > [Agent] > Overview — System Inventory section
- **Processes:** IT Hygiene > Processes — all running processes with PIDs, user, CPU, and memory usage
- **Ports:** IT Hygiene > Ports — all open TCP/UDP ports — identify any unexpected listening services
- **Packages:** IT Hygiene > Packages — all installed software with versions — cross-reference with vulnerability findings
- **Hardware:** CPU model, RAM, and storage information for asset management

6.9 Compliance Dashboards

Built-in dashboards for PCI DSS, GDPR, HIPAA, NIST 800-53, and TSC compliance frameworks. Wazuh includes pre-built compliance dashboards that automatically tag security events against regulatory requirements.

GDPR Dashboard

- **Navigate to:** Dashboard > Overview > GDPR tile
- **What it shows:** Maps Wazuh alerts to specific GDPR articles — e.g. Article 25 (data protection by design), Article 32 (security of processing)
- **Audit evidence:** Export GDPR alert reports to use as evidence in a GDPR audit

NIST 800-53 Dashboard

- **Navigate to:** Dashboard > Overview > NIST 800-53 tile
- **Use:** Maps detected events to NIST controls — useful for ISO 27001 alignment

6.10 Cloud Security

Monitors AWS, Google Cloud, Office 365, and GitHub security events directly via their native APIs, without requiring an agent on cloud resources.

AWS Integration

```
# In /var/ossec/etc/ossec.conf on the Manager:
<wodle name="aws-s3">
  <disabled>no</disabled>
  <interval>10m</interval>
  <bucket type="cloudtrail">
    <name>your-cloudtrail-bucket</name>
    <aws_profile>default</aws_profile>
    <path>AWSLogs/ACCOUNT_ID/CloudTrail/</path>
```

```
</bucket>
</wodle>
```

Office 365 Integration

- **Navigate to:** Dashboard > Overview > Office 365 tile
- **What it shows:** User login events, file access, Teams messages, admin actions from O365 audit logs
- **Use case:** Monitor for suspicious login locations or bulk data download from OneDrive

7. Log Monitoring Guide

This section provides a practical guide to monitoring logs across all endpoints using the Wazuh dashboard. Regular log monitoring is essential for detecting security incidents early and maintaining compliance.

7.1 Daily Monitoring Routine

The following steps should be performed each morning to check overnight activity:

- Open Wazuh Dashboard and navigate to Overview
- Check the '24 Hours Alerts' panel — focus on Critical (15+) and High (12–14) alerts first
- If any Critical alerts exist, click through to investigate the specific agent and event immediately
- Check Agents Summary — confirm all agents show status 'active'. Any disconnected agents require investigation.
- Review the MITRE ATT&CK overview — note any new tactics not seen previously
- Check Vulnerability Detection for any new Critical CVEs published overnight

7.2 Investigating a Specific Alert

- Navigate to Threat Hunting and set the time window to the period of interest
- Use filter: rule.level: [12 TO 15] to show only high/critical events
- Click an event to expand it — review: agent.name, data.srcip, rule.description, full.log
- Check if the source IP is internal or external — external IPs should be checked against threat intelligence
- If confirmed malicious: isolate the endpoint, preserve logs, and follow the incident response procedure

7.3 Custom Log Collection

To collect logs from a specific application, add log collection config to the agent:

```
# In /var/ossec/etc/ossec.conf on the AGENT:
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/your-app.log</location>
</localfile>
```

```
# For JSON logs:
<localfile>
  <log_format>json</log_format>
  <location>/var/log/app/*.json</location>
</localfile>

sudo systemctl restart wazuh-agent
```

7.4 Creating Custom Alert Rules

Custom rules allow the security team to define organisation-specific detections. Rules are stored on the manager:

```
# Custom rules file: /var/ossec/etc/rules/local_rules.xml
<group name="local,custom,">
  <!-- Alert on after-hours logins -->
  <rule id="100001" level="10">
    <if_sid>18107</if_sid>
    <time>22:00 - 06:00</time>
    <description>SSH login outside business hours</description>
    <mitre><id>T1078</id></mitre>
  </rule>

  <!-- Alert on sudo usage -->
  <rule id="100002" level="8">
    <if_sid>2502</if_sid>
    <description>Sudo command used on endpoint</description>
  </rule>
</group>

# Test rule syntax:
sudo /var/ossec/bin/wazuh-logtest

# Restart manager to load new rules:
sudo systemctl restart wazuh-manager
```

7.5 Generating Compliance Reports

- Navigate to Wazuh Dashboard > Reporting (or the relevant compliance dashboard e.g. GDPR)
- Set the date range for the report period
- Select the agents to include (all agents for a full network report)
- Click 'Generate Report' — Wazuh creates a PDF report suitable for audit purposes
- Download and store the report securely for compliance evidence

i NOTE: For GDPR compliance, generate monthly Wazuh GDPR compliance reports and retain them for a minimum of 12 months as evidence of continuous security monitoring.

8. Agent Maintenance & Management

8.1 Checking Agent Status

```
# Check agent service status (on endpoint):
sudo systemctl status wazuh-agent

# Check agent connectivity from manager:
sudo /var/ossec/bin/agent_control -l

# Restart a specific agent remotely from manager:
sudo /var/ossec/bin/agent_control -R <agent_id>
```

8.2 Updating Agent Configuration

```
# The main config file on Ubuntu agents:
sudo nano /var/ossec/etc/ossec.conf

# After any config change, restart:
sudo systemctl restart wazuh-agent

# View agent logs for troubleshooting:
sudo tail -f /var/ossec/logs/ossec.log
```

8.3 Upgrading Agents

```
# Upgrade all Linux agents from the manager:
sudo /var/ossec/bin/agent_upgrade -a

# Or upgrade individual agent:
sudo /var/ossec/bin/agent_upgrade -i <agent_id>
```

9. GDPR Article 32 Alignment

GDPR Article 32 requires organisations to implement appropriate technical measures to ensure a level of security appropriate to the risk. This Wazuh SIEM deployment directly supports that requirement through continuous endpoint monitoring and automated compliance reporting.

GDPR Requirement	How Wazuh Addresses It
Art. 32 — Appropriate technical measures	Continuous monitoring of all endpoints via SIEM with real-time alerting on security events
Art. 32 — Ongoing security testing	SCA benchmark scoring (CIS, PCI DSS, NIST) with pass/fail tracking and remediation guidance
Art. 32 — Confidentiality & integrity	FIM monitors all critical directories for unauthorised file changes in real time
Art. 25 — Data protection by design	Vulnerability detection continuously surfaces unpatched CVEs for proactive remediation
Art. 5 — Accountability	GDPR compliance dashboard generates audit-ready reports

	with event-to-article mapping
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10. Summary

Deployment Goal	SIEM coverage across all endpoints — real-time threat detection & GDPR compliance monitoring
Wazuh Version	v4.14.5
Agents Deployed	18 agents
Key Features Used	Threat Hunting · FIM · SCA · Vulnerability Detection · MITRE ATT&CK · Compliance Dashboards
Compliance	GDPR Article 32 — Continuous endpoint security monitoring & audit trail
Custom Rules	After-hours login detection, sudo usage monitoring
Status	All 18 agents active and reporting — successfully deployed and verified
Completed By	Karthikeyan T

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